Section 3.6: Quantified Statements

We are now prepared to use our tools for working with sets to work with statements like "all frogs are green" or "no bears are funny," which we would like to use in arguments.

Concept 1 – Quantified Statements

Definition: Quantified Statements
A **quantified statement** is a statement about **all** or **some** members of a group. In other words, it is a statement about **all** or **some** elements of a set.

The following are examples of quantified statements:

- All runners are athletes.
- Some dogs have fleas.
- Some doctors do not play golf.
- No math teachers are funny.

Quantified statements tell us that _____ or _____ of the elements of a _____ are or are not in another _____. To help us understand—and ultimately symbolize—quantified statements, we will use Venn diagrams as well as another kind of diagram, an Euler diagram.

Definition: Euler Diagram
An **Euler diagram** (pronounced *oiler*) is a diagram similar to a Venn diagram, but in which subsets that are known to be empty are removed.

We will build Euler diagrams based on various Venn diagrams throughout this section.

Concept 2 – Universal Statements

Definition: Universal Statement
A **universal statement** tells us something about _____ of the elements of a set.
A **positive universal** tells us that all the elements of a set _____ belong to another set.
A **negative universal** tells us that all the elements of a set _____ belong to another set; in other words, _____ of the elements of one set also belong to another.

Example 1: The statement below is a _____ universal statement.

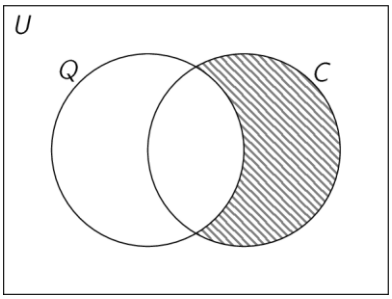
All cats are quick.

If we define the sets:

$C =$ the set of all _____

$Q =$ the set of all _____

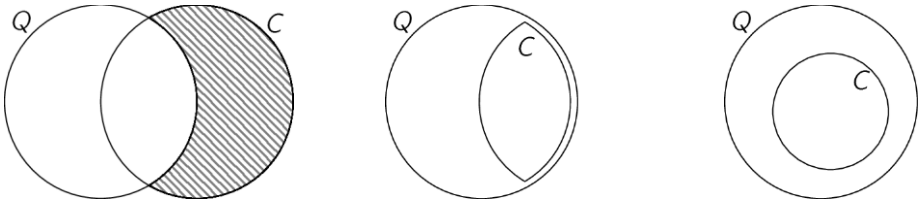
we get the following Venn diagram, where the shaded region indicates a set that we know is empty.



To convert this to an Euler diagram, we follow two steps:

1. Stop drawing the universal set (it's assumed in an Euler diagram).
2. Delete the parts that are known to be empty.
3. Redraw so that each set is represented by a circle.

For this Euler diagram, here is the process.



We will think of the Euler diagram (the last one in the sequence) as representing the statement "All cats are quick." Notice that the set of cats is _____ the set of quick things—in other words, cats are a _____ of quick things.

Example 2: The statement below is a _____ universal statement.

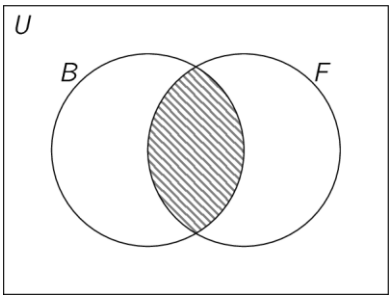
No birds are fish.

If we define the sets:

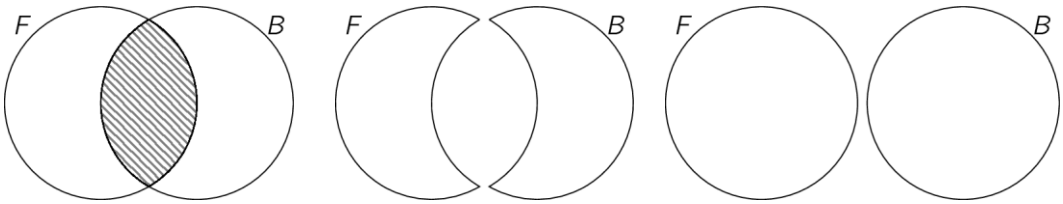
$B =$ the set of all _____

$F =$ the set of all _____

we get the following Venn diagram, where the shaded region indicates a set that we know is empty.



We can follow the same steps as before to convert this to an Euler diagram.



We will think of the Euler diagram (the last one in the sequence) as representing the statement "No birds are fish." Notice that the set of birds and the set of fish do not _____—in other words, the sets are _____ .

Concept 3 – Existential Statements

Definition: Existential Statement
An **existential statement** tells us something about _____ of the elements of a set. A positive existential tells us that an element of one set _____ belong to another set. A negative universal tells us that an element of one set _____ belong to another set.

We use the word "some" when we make an existential statement. Note that, when we're speaking formally, "some" means "at least one." In particular:

- "Some" does not mean _____
- "Some" does not mean _____
  (that is, for example, "some frogs are green" and "all frogs are green" can _____ be true).

Example 3: The statement below is a _____ existential statement.

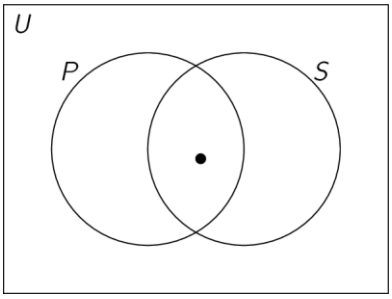
Some people can sing

If we define the sets:

$P =$ the set of all _____

$S =$ the set of all _____

then we can draw the Venn diagram below:



The dot represents the fact that there *must* be something in the overlap between the two circles.

Example 4: The statement below is a _____ existential statement.

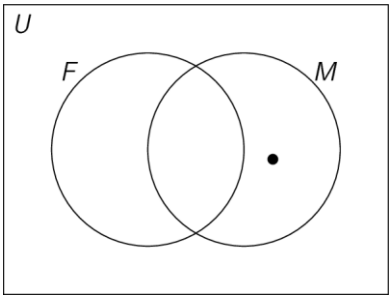
Some math teachers are not funny.

If we define the sets:

$M =$ the set of all _____

$F =$ the set of all _____

then we can draw the Venn diagram below:



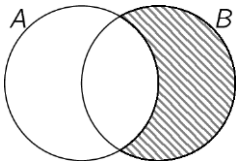
The dot represents the fact that there *must* be something in the overlap between the two circles.

Concept 4 – Possible Euler Diagrams

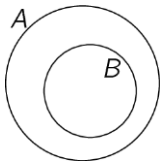
So far, we've seen two possible Euler diagrams. For sets A and B, we have seen:

- "All [set B] are [set A]"

Venn diagram

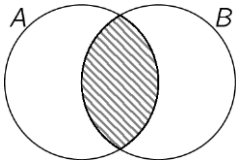


Euler diagram

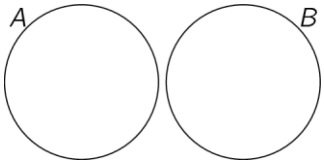


- "No [set A] are [set B]" or "No [set B] are [set A]"

Venn diagram



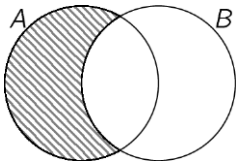
Euler diagram



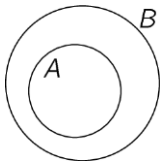
There are three other Euler diagrams that we might draw.

- "All [set A] are [set B]"

Venn diagram



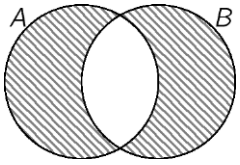
Euler diagram



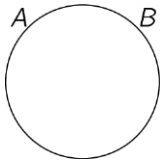
This is like the first one, but in the opposite _____

- "All [set A] are [set B]" *and* "All [set B] are [set A]" at the same time.

Venn diagram



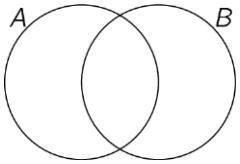
Euler diagram



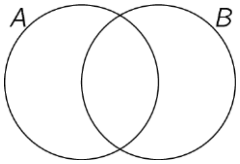
Here, the two sets are the _____ set, so they're the same _____ .

- The last one is the default diagram, with all possible overlaps. So there are some things that are in both sets, some that are only in A, some that are only in B.

Venn diagram



Euler diagram



Concept 5 – Negations of Quantified Statements

Let's consider how to negate a quantified statement.

Example 5: Negate the statement below.

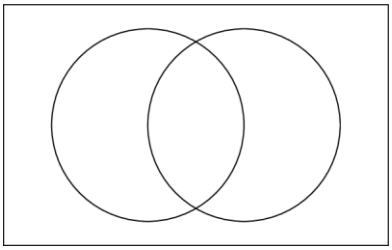
All trees are tall.

If we define the sets:

$R =$ the set of all _____

$L =$ the set of all _____

We'll use the Venn diagram below to think through the negation.



- When the statement is *true*, _____
  _____
- So when the statement is *false* (that is, if the negation is true), _____
  _____

The negation of "All trees are tall." is _____
_____

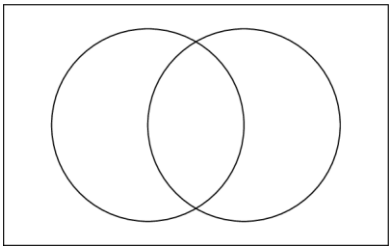
Example 6: Negate the statement below.

Some artists use oil paints.

If we define the sets:

$A =$ the set of all _____

$O =$ the set of all _____



The negation of "Some artists use oil paints" is _____
_____

You do not need to memorize how to negate a quantified statement! Instead, draw (or imagine) the Venn diagram, and think through what the negation must be.

Concept 6 – Symbolizing a Positive Universal

In the context of an argument, we will be applying a universal statement to a _____.

Example 7: In the argument below, how could you symbolize the universal statement?

All runners are athletes. Lydia is a runner. Therefore, Lydia is an athlete.

This argument has two premises and a conclusion.

Premise 1: _____

Premise 2: _____

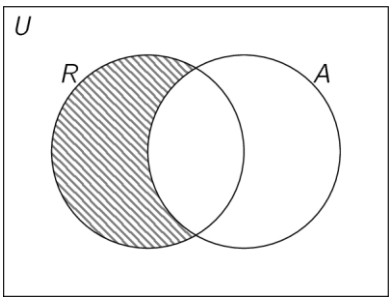
Conclusion: _____

To symbolize premise _____ and the conclusion, we need two simple statements:

$p =$ _____

$q =$ _____

We hope that we can symbolize premise _____ in terms of these two simple statements, too. To do this, we'll use both a truth table and a Venn diagram!



Row	p	q	
1	T	T	
2	T	F	
3	F	T	
4	F	F	

Looking at the for rows of the truth table, this is what we see.

1. $p =$ T and $q =$ T, so _____

   _____

   This (can | can't) be true if "all runners are athletes".
2. $p =$ T and $q =$ F, so _____

   _____

   This (can | can't) be true if "all runners are athletes".
3. $p =$ F and $q =$ T, so _____

   _____

   This (can | can't) be true if "all runners are athletes".
4. $p =$ F and $q =$ F, so _____

   _____

   This (can | can't) be true if "all runners are athletes".

Only row _____ can't be true all runners are athletes; this is _____

We'll use a _____ to symbolize a positive universal.

Concept 7 – Symbolizing a Negative Universal

Example 8: In the argument below, how could you symbolize the universal statement?

No math teachers are nice. Bob is a math teacher. Therefore, Bob is not nice.

This argument has two premises and a conclusion.

Premise 1: _____

Premise 2: _____

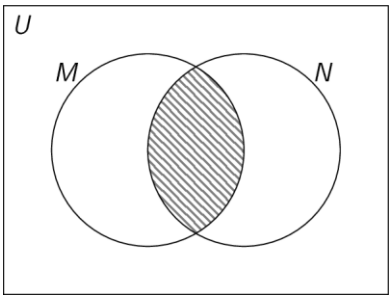
Conclusion: _____

To symbolize premise _____ and the conclusion, we need two simple statements:

$p =$ _____

$q =$ _____

We hope that we can symbolize premise _____ in terms of these two simple statements, too. To do this, we'll use both a truth table and a Venn diagram!



Row	p	q	
1	T	T	
2	T	F	
3	F	T	
4	F	F	

Looking at the for rows of the truth table, this is what we see.

1. $p =$ T and $q =$ T, so _____

   _____

   This (can | can't) be true if "no math teachers are nice".
2. $p =$ T and $q =$ F, so _____

   _____

   This (can | can't) be true if "no math teachers are nice".
3. $p =$ F and $q =$ T, so _____

   _____

   This (can | can't) be true if "no math teachers are nice".
4. $p =$ F and $q =$ F, so _____

   _____

   This (can | can't) be true if "no math teachers are nice".

Only row _____ can't be true all runners are athletes; we haven't seen this truth table before!

We're trying to symbolize a truth table with _____ "true" rows. Since order doesn't matter, we'll try to use _____.

p	q	Goal	$\sim p$	$\sim q$	$p \vee q$	$\sim p \vee q$	$p \vee \sim q$	$\sim p \vee \sim q$

The statement we're trying to symbolize is _____.

We'll use a _____with _____ to symbolize a negative universal.